

The Levitron Lab: An Interactive Curriculum for Magnetic Levitation Physics

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We describe *The Levitron Lab*, a browser-based collection of interactive simulations that teach how different magnetic levitation toys work by climbing a dimensionality ladder from Earnshaw’s theorem to full rigid-body and driven-rotor dynamics. Each of five modules foregrounds a live, animated levitating object with sliders on every physical parameter, diagnostic plots, and inline equations. Feedback modules implement full PID control with live Jacobian eigenvalues; the gyroscopic module implements the symmetric-top nutation analysis from the author’s MIT 8.223 project. The tool is open source, runs offline, and deploys in one click via GitHub Codespaces (<https://github.com/saldajani/levitron-lab>). Supplementary interactive materials are available at the repository and in the accompanying video.

I. INTRODUCTION

Earnshaw’s theorem forbids stable equilibrium for a magnetic dipole in a static field: wherever forces balance, the potential is a saddle³. Yet levitating globes, spinning tops, and motor-driven floaters are sold as toys. Each evades the theorem by a distinct mechanism: time-varying feedback fields; gyroscopic spin; or frequency-locked rotation in a driven field^{4–6}.

The Levitron Lab presents all three escape routes in one curriculum, ordered by *which directions are unstable*—a classification articulated in conversation with an experimental atomic physicist who keeps a classic Levitron on his desk as a tabletop model of a magnetic atom trap¹⁰.

II. EARNSHAW’S THEOREM AND THE THREE ESCAPE ROUTES

Module 0 places a free magnet above a fixed magnet. The user hunts for $F(z) = 0$ and discovers $\partial^2 U / \partial z^2 < 0$: passive levitation is impossible (Fig. 1).

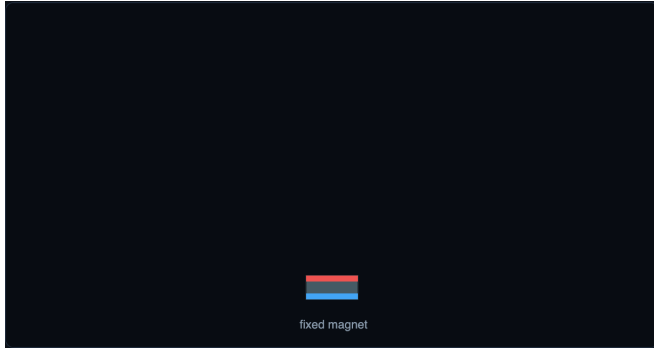


FIG. 1 Earnshaw primer: the free magnet cannot rest at the force balance — an unstable hilltop, not a bowl.

The three escape routes previewed are: (1) active feedback (Modules 1–2); (2) gyroscopic spin (Module 3); (3) driven rotation (Module 4).

III. THE FIVE MODULES AND IMPLEMENTATION

The application is built in React, TypeScript, and **three.js**. Physics integrates on a fixed 1 ms timestep decoupled from rendering; feedback modules use semi-implicit Euler; rigid-body modules use RK4.

III.A. One- and two-dimensional feedback

Module 1 models the floating-globe class: stable horizontally, unstable in z , corrected by PID control

$$I = \text{clamp}\left(K_p e + K_i \int e dt + K_d \dot{e}, 0, I_{\max}\right). \quad (1)$$

Module 2 is the mathematically distinct sibling: stable in z , unstable in x and y , corrected by four horizontal coils. Both modules compute the closed-loop Jacobian and display eigenvalues live (Fig. 2).



FIG. 2 Closed-loop eigenvalues for Module 1 (z instability tamed) and Module 2 (x, y instabilities tamed). Open-loop unstable roots cross $\text{Re } \lambda = 0$ when detuned.

Thermal drift with $K_i = 0$ produces visible steady-state sag below setpoint; raising K_i eliminates the offset — a teaching point tied to real levitating-globe trim pots.

III.B. Dynamical rotor levitation

Module 4 integrates a floater in a motor-driven rotor field^{5–7}. Trapping onset is governed by rotor geometry; the module displays $U_f(z_f)$ morphing as the lateral-displacement parameter crosses its critical value (Fig. 3).

IV. THE GYROSCOPIC TOP AND THE 8.223 SYMMETRIC-TOP ANALYSIS

The classic Levitron is stabilized by angular momentum and magnetic force alone^{3,4}. Module 3 implements

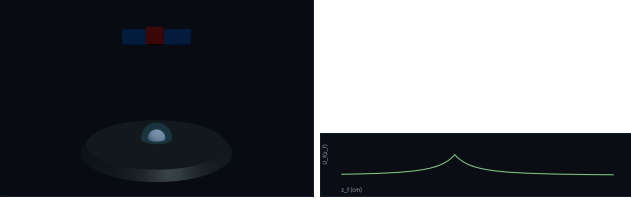


FIG. 3 Dynamical rotor module: frequency-locked floater (left) and floater potential $U_f(z_f)$ (right).

the symmetric-top mechanics developed in the author's MIT 8.223 project^{1,2}.

IV.A. Lagrangian, conserved momenta, and precession roots

For a symmetric top with Euler angles (φ, θ, ψ) (precession, nutation, spin), the conserved momenta are

$$p_\psi = I_s(\dot{\psi} + \dot{\varphi} \cos \theta), \quad (2)$$

$$p_\varphi = (I_p \sin^2 \theta + I_s \cos^2 \theta) \dot{\varphi} + I_s \dot{\psi} \cos \theta. \quad (3)$$

At fixed θ , there are slow and fast precession roots; the slow root satisfies $\dot{\varphi}_{\text{slow}} = mgl/p_\psi$. Launching at the slow-precession equilibrium with $\dot{\theta} = 0$ and $\cos \theta_{\text{stable}} = p_\varphi/p_\psi$ is essential — arbitrary transverse angular velocity at $\theta = 0$ ejects the top immediately.

IV.B. Restoring potential and bounded nutation

Nutation satisfies an effective one-dimensional equation with potential

$$U_{\text{res}}(\theta) = \frac{p_\psi^2}{2I_s} + \frac{(p_\varphi - p_\psi \cos \theta)^2}{2I_p \sin^2 \theta} + V_{\text{eff}}(\theta), \quad (4)$$

where V_{eff} combines gravity and magnetic upright bias (replacing $mgl \cos \theta$ in the gravitational-top form). Bounded nutation requires $\theta_{\min} < \theta < \theta_{\max}$ at the turning points of U_{res} (Fig. ??). The device “works only when nice and upright”: the allowable cone half-angle θ_c shrinks as spin drops toward $\omega_{\min}$ or as the field weakens with temperature.

The spin-rate window ($\omega_{\min} \approx 19$ rps for ferrite) and nutation cone are *both* required: the window ensures adiabatic following of the field (Berry phase⁴); the cone keeps the axis upright enough for that following to hold. Startup diagnostics verify $p_\psi^2 \gg 4mglI_p \cos \theta$ and that U_{res} possesses a local minimum before integration begins.

V. FROM TOY TO ATOM TRAP

A neutral atom with spin angular momentum is a microscopic gyroscope; magnetic trapping uses the same adiabatic-invariant logic as the Levitron^{4,10}. The tool includes a dedicated page drawing this correspondence explicitly.

VI. CLASSROOM USE AND AVAILABILITY

The Levitron Lab runs fully client-side with no backend. GitHub Codespaces provides one-click deployment: `npm install` and `npm run dev -- --host start` automatically. Figure capture for papers uses `npm run figures` (Playwright). See <https://github.com/saldajani/levitron-lab>.

VII. CONCLUSION

By rendering the levitating body alongside the equations that govern it, and by ordering devices according to unstable dimensionality, The Levitron Lab turns familiar toys into a unified quantitative curriculum from Earnshaw's prohibition to atom-trap analogies.

ACKNOWLEDGMENTS

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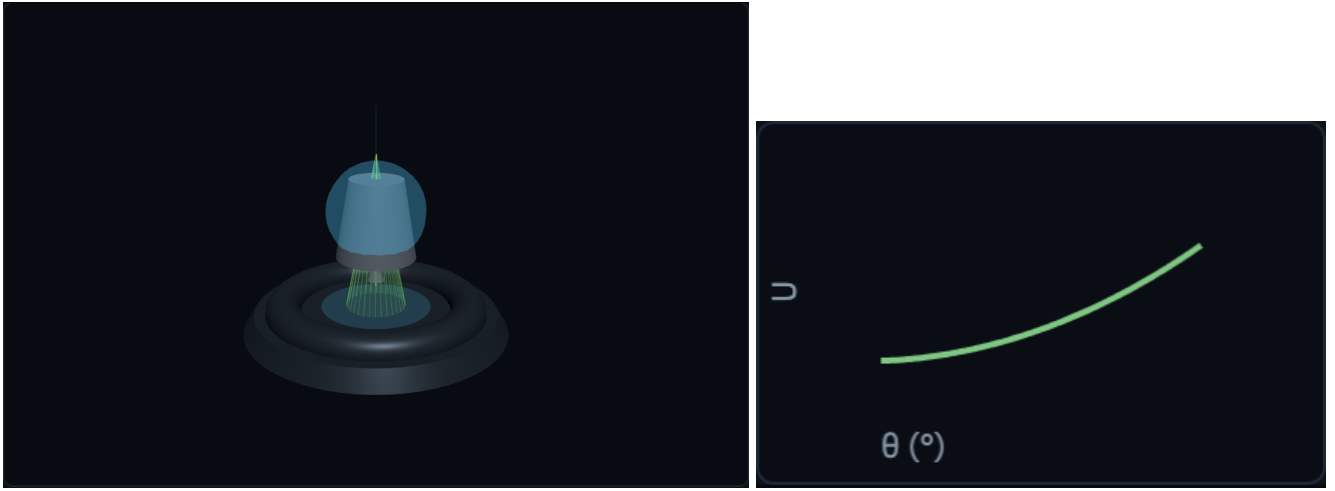


FIG. 4 (Left) Stable gyroscopic Levitron with nutation cone and symmetry axis. (Right) $U_{\text{res}}(\theta)$ showing the restoring well; $\theta_{\min} < \theta < \theta_{\max}$.

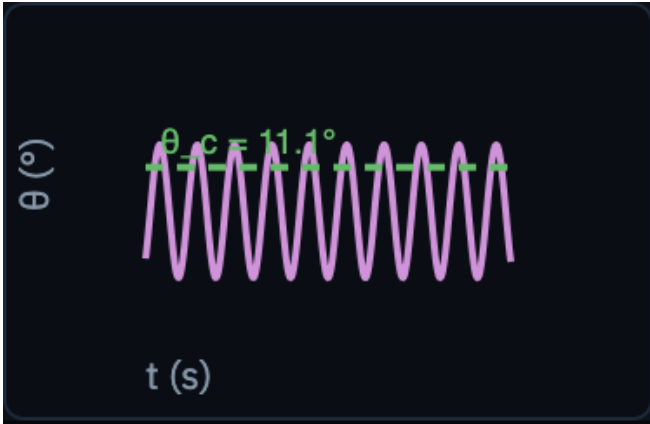


FIG. 5 Nutation angle $\theta(t)$: bounded nodding when stable; diverging growth before crash.